

Lecture 13 - Double/debiased Machine Learning

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Applications of Shrinkage Methods

- Lecture 12: group-specific mean, $\hat{y}_v^\theta = \theta \bar{y} + (1 - \theta) \bar{y}_v$, or using Ridge.
- EJR example: using Lasso for variable selection – identifying keywords that are most predictive of each gender.
- Variable selection is particularly useful when we are dealing with high-dimensional data, especially if the number of features, p exceeds the number of observations N .

Double/debiased Machine Learning

- Today we will introduce double/debiased machine learning for treatment and structural parameters (Chernozhukov et al. 2018 Journal of Econometrics).
 - Low-dimensional parameter of interest, θ_0 : e.g., ATE, ATT, LATE
 - High-dimensional nuisance parameters, η_0 : e.g., lots of features in X , which may be interacted with each other. One may employ regularization to select features/reduce variance.
- Chernozhukov et al. proposes we use regularization **twice**. Key ideas we will revisit in this framework:
 - Frisch-Waugh Theorem
 - Regularization
 - Cross validation (cross-fitting)

Review: Frisch-Waugh Theorem

- Consider the following regression of an outcome Y on a treatment variable D , and controls X :

$$Y = D\theta_0 + X\beta + \epsilon$$

in which $\epsilon \perp (D, X)$

- We are interested in θ_0 , and by Frisch-Waugh,

$$\theta_0 = \frac{E[\tilde{D}Y]}{E[\tilde{D}^2]}$$

in which $\tilde{D} := D - E^*[D|X]$ represents the residual from the auxiliary regression of D on X :

$$D = X\Gamma + \tilde{D}$$

Review: Frisch-Waugh Theorem

- What if $|X| \uparrow \uparrow$? Or interactions between covariates (or even between X and D) matter? Regularization may help us reduce variance in our estimators, but we want to keep the bias small, and get a credible estimate of low-dimensional θ_0 .
- The key idea is to use ML twice: in the outcome regression, and in the auxiliary regression!

Motivating Framework: Partially Linear Model

- Y : outcome, D : exogenous treatment variable, $X = (X_1, \dots, X_p)$: controls/covariates.

$$Y = D\theta_0 + g_0(X) + U, \quad E[U|X, D] = 0 \quad (1)$$

$$D = m_0(X) + V, \quad E[V|X] = 0 \quad (2)$$

- θ_0 : treatment effect.
- The confounding factors X affect outcome via the function $g_0(X)$, and treatment D via $m_0(X) \sim "X\beta"$ in the auxiliary regression.
- We want to use regularization to form a ML estimator: $D\hat{\theta}_0 + \hat{g}_0(X)$.

Idea 1: Use ML to estimate $g_0(X)$

- Suppose in a training dataset, we obtain $\hat{g}_0(X)$ by using Lasso or other ML methods. Consider the following estimate of $\hat{\theta}_0$ given $\hat{g}_0$:

$$\hat{\theta}_0 = \left(\frac{1}{n} \sum_{i \in \text{test}} D_i^2 \right)^{-1} \frac{1}{n} \sum_{i \in \text{test}} D_i (Y_i - \hat{g}_0(X_i))$$

$$\text{plug in } Y_i = D_i \theta_0 + g_0(X_i) + U_i$$

$$\Rightarrow \hat{\theta}_0 = \theta_0 + \left(\frac{1}{n} \sum_{i \in \text{test}} D_i^2 \right)^{-1} \frac{1}{n} \sum_{i \in \text{test}} D_i \underbrace{(g_0(X_i) - \hat{g}_0(X_i))}_{\text{regularization bias}} + U_i$$

- The estimation error in $\hat{\theta}_0$ has two parts: " $D_i U_i$ ", and " $D_i(g_0 - \hat{g}_0)$ ", which is subject to regularization bias.

$$\begin{aligned} \sqrt{n}(\hat{\theta}_0 - \theta_0) &= \left(\frac{1}{n} \sum_{i \in \text{test}} D_i^2 \right)^{-1} \underbrace{\frac{1}{\sqrt{n}} \sum_{i \in \text{test}} D_i U_i}_{\xrightarrow{p} 0} \\ &\quad + \left(\frac{1}{n} \sum_{i \in \text{test}} D_i^2 \right)^{-1} \frac{1}{n} \sum_{i \in \text{test}} D_i (g_0(X_i) - \hat{g}_0(X_i)) \end{aligned}$$

Idea 2: Orthogonalization

- Chernozhukov et al. employs an orthogonalized formulation to overcome regularization bias. In the training set, partial out the effect of X on D and obtain the orthogonalized regressor

$$\hat{V} = D - \hat{m}_0(X).$$

And in the training set estimate $\hat{g}_0(X)$ in the outcome regression. So we are doing double ML predictions: Y given X , and D given X !

- The idea of partialling the effect of X out from D is just like getting the residual from the auxiliary regression in Frisch-Waugh.

Double/Debiased ML Estimator - Regularization Bias

- Under double ML (“DML”), the estimator for θ_0 ,

$$\tilde{\theta}_0 = \left(\frac{1}{n} \sum_{i \in \text{test}} \hat{V}_i^2 \right)^{-1} \left(\frac{1}{n} \sum_{i \in \text{test}} \hat{V}_i (Y_i - \hat{g}_0(X_i)) \right)$$

plug in $Y_i = D_i \theta_0 + g_0(X_i) + U_i$, and $D_i = \hat{V}_i + \hat{m}_0(X_i)$, we have:

$$\tilde{\theta}_0 - \theta_0 = \left(\frac{1}{n} \sum_{i \in \text{test}} \hat{V}_i^2 \right)^{-1} \underbrace{\frac{1}{n} \sum_{i \in \text{test}} \hat{V}_i (g_0(X_i) - \hat{g}_0(X_i) + U_i)}_{(*)}$$

Double/Debiased ML Estimator - Regularization Bias

- Focus on $(*)$: note $\hat{V}_i = D_i - \hat{m}_0(X_i) = V_i + m_0(X_i) - \hat{m}_0(X_i)$,

$$\begin{aligned} (*) &= \frac{1}{n} \sum_i (V_i + m_0(X_i) - \hat{m}_0(X_i)) \times (g_0(X_i) - \hat{g}_0(X_i) + U_i) \\ &= \underbrace{\frac{1}{n} \sum_i V_i U_i}_{\text{ok}} \\ &\quad + \underbrace{\frac{1}{n} \sum_i (m_0(X_i) - \hat{m}_0(X_i))(g_0(X_i) - \hat{g}_0(X_i))}_{\text{regularization bias}} \\ &\quad + \underbrace{\frac{1}{n} \sum_i (m_0(X_i) - \hat{m}_0(X_i)) U_i + \frac{1}{n} \sum_i V_i (g_0(X_i) - \hat{g}_0(X_i))}_{\text{in-sample/overfitting bias}} \end{aligned}$$

the regularization bias in g_0 and m_0 depends on the **product** of the estimation errors in $\hat{g}_0, \hat{m}_0$. It can vanish even when both functions are estimated at relatively slow rates. “Double/debiased” through applying ML twice (2 chances to get it right!)

Double/Debiased ML Estimator - Regularization Bias

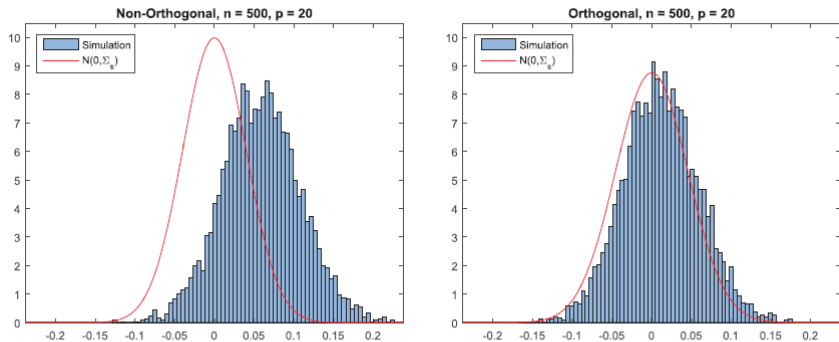


Figure 1. Comparison of the conventional and double ML estimators. [Colour figure can be viewed at wileyonlinelibrary.com]

Why Training-Test Split or Cross Fitting?

- So far, we have assumed $\hat{g}_0(X)$, $\hat{m}_0(X)$ are estimated in a training set, while $\tilde{\theta}_0$ is estimated in a holdout/test set.
- What happens if we estimate all of them in the same sample?
Overfitting Bias!
- Note in (*), the 3rd component

$$\frac{1}{n} \sum_i (m_0(X_i) - \hat{m}_0(X_i)) U_i + \frac{1}{n} \sum_i V_i (g_0(X_i) - \hat{g}_0(X_i))$$

would not vanish if there is overfitting in sample.

Why Training-Test Split or Cross Fitting?

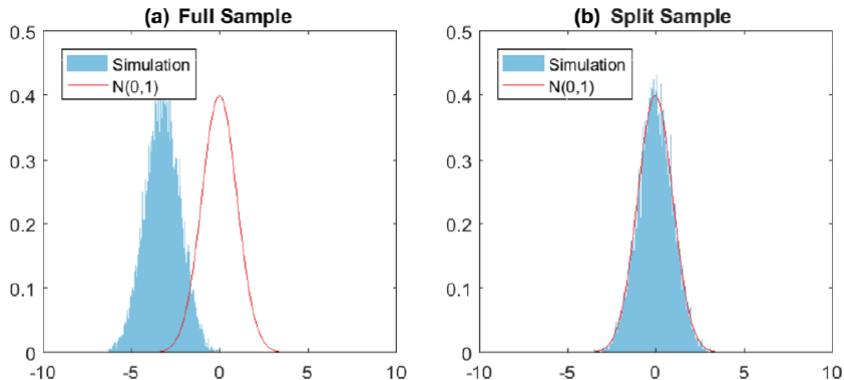


Figure 2. Comparison of full-sample and cross-fitting procedures. [Colour figure can be viewed at wileyonlinelibrary.com]

Cross Fitting Procedure

- 1 Take a K -fold random partition of our data: $(I_k)_{k=1}^K$ of observations $\{1 \dots N\}$ such that each fold has N/K observations. Define leave- k -out as $I_k^c = \{1 \dots N\} \setminus I_k$.
- 2 For each k , estimate $\hat{m}_{0,k}, \hat{g}_{0,k}$ in the leave- k -out sample I_k^c via regularization. Then estimate $\tilde{\theta}_{0k}$ in the test sample I_k .
- 3 Aggregate the estimators:

$$\tilde{\theta}_0 = \frac{1}{K} \sum_{k=1}^K \tilde{\theta}_{0k}.$$

Question: what is $K = N$ -fold cross fitting?

Choice of K : Bias-Variance trade-off again

- Small K s.t. $K = 2$: in each fold the size of the training set is

$$\frac{(K-1)}{K} \times N,$$

which is smaller for smaller K . The estimators $\hat{m}_{0,k}, \hat{g}_{0,k}$ are noisier. Meanwhile, there is lower risk of overfitting.

- Large $K = 4, 5$ work better than $K = 2$: smaller variance given larger training samples in each fold.
- If $K = N$, equivalent to leave-1-out. High risk of overfitting.

Application 1: Partially Linear IV

- Consider a partially linear IV model:

$$\begin{aligned} Y &= D\theta_0 + g_0(X) + U, & E[U|X, Z] &= 0 \\ Z &= m_0(X) + V, & E[V|X] &= 0 \end{aligned} \quad (3)$$

- Z is the instrument variable for endogenous treatment variable D .
What makes Z a valid instrument?
 - First stage/relevance: $E[D|Z, X] \neq E[D|X]$.
 - Exclusion restriction: $E[U|Z] = 0$.
- Under $E[U|X, Z] = 0$, express the exclusion restriction as:

$$E[UV] = 0 = E[(Y - D\theta_0 - g_0(X)) \times (Z - m(X))] \quad (4)$$

the nuisance parameters in (4) are g, m , which are solved in training sets.

Application 1: Partially Linear IV

- Alternatively, residualize all Y, D, Z w.r.t. X (Robinson-style):

$$\begin{aligned}l_0(X) &= E[Y|X], \quad r_0(X) = E[D|X] \\ m_0(X) &= E[Z|X]\end{aligned}$$

then

$$\begin{aligned}Y &= D\theta_0 + g_0(X) + U \\ &= (r_0(X) + \tilde{D})\theta_0 + g_0(X) + U \\ &= \theta_0\tilde{D} + \underbrace{r_0(X)\theta_0 + g_0(X)}_{l_0(X)} + U\end{aligned}$$

in which $r_0(X)\theta_0 + g_0(X) = l_0(X)$ as $(\tilde{D}, U)$ are mean independent of X .

Application 1: Partially Linear IV

- Plug the conditional means into (4):

$$E[(Y - l_0(X) - \theta_0(D - r_0(X))) \times (Z - m_0(X))] = 0 \quad (5)$$

It's easier to obtain the DML estimators based on (5), given that the nuisance parameters are conditional means – l_0, r_0, m_0 .

- 1 K -fold random partition of our data: $(I_k)_{k=1}^K$
- 2 In each fold k of the cross fitting, first estimate l_{0k}, r_{0k}, m_{0k} in the leave- k -out training sample, I_k^c . Then in the test set I_k , look for $\tilde{\theta}_{0k}$ that solve (5) (method of moments).
- 3 Aggregate the estimators $\tilde{\theta}_0 = \frac{1}{K} \sum \tilde{\theta}_{0k}$.

Application 2: Average Treatment Effects

- Recall we could estimate ATE under the **unconfoundedness** assumption (aka conditional independence/selection by observables) in Rosenbaum and Rubin (1983):

$$(Y_i(0), Y_i(1)) \perp D_i \mid X_i.$$

We introduce two ways to estimate ATE:

- 1 Estimate CATE first conditional on each possible value of X :

$$\begin{aligned} E[Y_i | D_i = 1, X] &= E[Y_i(1) | D_i = 1, X] = E[Y_i(1) | X] \\ \text{CATE}(X) &:= E[Y_i(1) - Y_i(0) | X] \\ &= E[Y_i | D_i = 1, X] - E[Y_i | D_i = 0, X] \\ \text{ATE} &:= E[Y_i(1) - Y_i(0)] \\ &= E_X[E[Y_i(1) - Y_i(0) | X]] \\ &= \int_x \text{CATE}(x) dF(x) \end{aligned}$$

Application 2: Average Treatment Effects

- ② Propensity score: the unconfoundedness assumption implies

$$(Y_i(0), Y_i(1)) \perp D_i \mid p(X)$$

in which $p(X) = E[D_i|X]$. Under the correctly specified propensity $p(X)$, we can show:

$$ATE = E\left[\frac{D_i Y_i}{p(X_i)}\right] - E\left[\frac{(1 - D_i) Y_i}{1 - p(X_i)}\right].$$

This estimator is easier to obtain when we have continuous/high-dimensional X .

Application 2: Average Treatment Effects

- Doubly Robust Estimator (Robins et al. 1994): we actually have 2 chances to get it right — outcome models, and propensity score models. Suppose the true conditional means are:

$$E[Y(0)|X] = g_0(0, X), \quad E[Y(1)|X] = g_0(1, X) \\ E[D|X] = p(X)$$

We have estimators g for g_0 and m for p . Consider the following estimator for ATE, $E[\theta]$:

$$\theta := (g(1, X) - g(0, X)) + \left(\frac{D(Y - g(1, X))}{m(X)} - \frac{(1 - D)(Y - g(0, X))}{1 - m(X)} \right) \\ E[\theta] = E[g(1, X) - g(0, X)] \\ + E\left[\frac{D(Y - g(1, X))}{m(X)}\right] - E\left[\frac{(1 - D)(Y - g(0, X))}{1 - m(X)}\right] \quad (6)$$

Application 2: Average Treatment Effects

- ④ If $m(X) = p(X) = E[D|X]$ we have a correctly specified propensity score model, then

$$\begin{aligned} E\left[\frac{D(Y - g(1, X))}{m(X)}\right] &= E_X\left[\frac{E[D(Y - g(1, X))|X]}{m(X)}\right] \\ &= E_X[E[Y - g(1, X)|D = 1, X] \frac{E[D|X]}{m(X)}] \\ &= E_X[E[Y(1) - g(1, X)|X]] \quad (7) \\ E\left[\frac{(1 - D)(Y - g(0, X))}{1 - m(X)}\right] &= E_X[E[Y(0) - g(0, X)|X]] \end{aligned}$$

So even if $g \neq g_0$ (outcome model misspecified), we still have:

$$\begin{aligned} E[\theta] &= E_X[E[g(1, X) - g(0, X)|X]] + E_X[E[Y(1) - g(1, X)|X]] - E_X[E[Y(0) - g(0, X)|X]] \\ &= E_X[E[Y(1) - Y(0)|X]] = E[Y(1) - Y(0)] = ATE! \end{aligned}$$

Application 2: Average Treatment Effects

- What if we have the right outcome model $g = g_0$ while $m(X) \neq E[D|X]$?

$$\begin{aligned} E[g(1, X) - g(0, X)] &= E[g_0(1, X) - g_0(0, X)] \\ &= E[E[Y(1) - Y(0)|X]] = ATE \\ E\left[\frac{D(Y - g(1, X))}{m(X)}\right] &= E_X\left[\underbrace{E[Y - g(1, X)|D = 1, X]}_{=0} \frac{E[D|X]}{m(X)}\right] \\ &= 0 = E\left[\frac{(1 - D)(Y - g(0, X))}{1 - m(X)}\right] \end{aligned}$$

So we also have $E[\theta] = ATE$ once outcome model $g = g_0$ is correct.

- If both g and m are mis-specified, we may get answers that are more wrong than using either method alone.

- Based on the doubly robust estimator above, Chernozhukov et al. proposes we solve:

$$\begin{aligned} E[\psi(\text{data}; \theta, g, m)] &= 0 & (8) \\ \text{for } \psi(\text{data}; \theta, g, m) &:= g(1, X) - g(0, X) \\ &+ \frac{D(Y - g(1, X))}{m(X)} - \frac{(1 - D)(Y - g(0, X))}{1 - m(X)} \\ &- \theta \end{aligned}$$

- To implement DML, again estimate g, m in training sets, plug them into ψ and solve (8).

Application 3: Local Average Treatment Effects

- Treatment effects could be heterogeneous. We spent a few lectures on local average treatment effects, which can be estimated if we have a valid instrument $Z \in \{0, 1\}$ for treatment $D \in \{0, 1\}$ that satisfies:
 - Z_i is relevant for D_i . $E[D] \neq E[D|Z = 1]$.
 - Random assignment: $Z_i \perp (D_i(0), D_i(1))$
 - Exclusion restriction: $Y_i(d, z) = Y_i(d)$, and $Z_i \perp (Y_i(0), Y_i(1))$.
- LATE = ATE for compliers with $D_i(1) > D_i(0)$ (under monotonicity $D_i(1) \geq D_i(0)$ for everyone, or no defiers). The Wald estimator for LATE is:

$$\frac{E[Y|Z = 1] - E[Y|Z = 0]}{E[D|Z = 1] - E[D|Z = 0]}$$

Application 3: Local Average Treatment Effects

We focused on the linear model, but the Wald estimator for LATE holds for nonlinear/interactive models too:

$$\begin{aligned}Y &= \mu_0(Z, X) + U, & E[U|Z, X] &= 0 \\D &= m_0(Z, X) + V, & E[V|Z, X] &= 0 \\Z &= r_0(X) + \zeta, & E[\zeta|X] &= 0 \\ \theta_0 &= \frac{E[\mu_0(1, X)] - E[\mu_0(0, X)]}{E[m_0(1, X)] - E[m_0(0, X)]}.\end{aligned}\tag{9}$$

- DML estimator for LATE is also doubly robust, and it solves " $\delta = \pi \times \theta$ ".

$$\begin{aligned}&\mu(1, X) - \mu(0, X) + \frac{Z(Y - \mu(1, X))}{r(X)} + \frac{(1 - Z)(Y - \mu(0, X))}{1 - r(X)} \\&- \theta \times (m(1, X) - m(0, X) + \frac{Z(D - m(1, X))}{r(X)} + \frac{(1 - Z)(D - m(0, X))}{1 - r(X)})\end{aligned}$$

that is, we are getting a minimum-distance estimator for LATE that corresponds to the WALD estimator if either (μ, m) or r is correctly specified.

Example: 401(k) on financial assets

- Section 6.2 applies DML-LATE to the question of the effect of 401(k) participation on financial assets. The instrument for 401(k) participation is eligibility, which is arguably exogenous after conditioning on income and job characteristics (Poterba et al. 1994).

Table 3. Estimated effect of 401(k) participation on net financial assets.

	Lasso	Reg. tree	Random forest	Boosting	Neural network	Ensemble	Best
LATE	8978	11073	11384	11329	11094	11119	10952
(twofold)	[2192] (3014)	[1749] (1849)	[1832] (1993)	[1666] (1718)	[1903] (2098)	[1653] (1689)	[1657] (1699)
LATE	8944	11459	11764	11133	11186	11173	11113
(fivefold)	[2259] (3307)	[1717] (1786)	[1788] (1893)	[1661] (1710)	[1795] (1890)	[1641] (1678)	[1645] (1675)

Note: Estimated LATE based on orthogonal estimating equations. Column labels denote the method used to estimate nuisance functions. Results are based on 100 splits with point estimates calculated the median method. The median standard errors across the splits are reported in brackets and standard errors calculated using the median method to adjust for variation across splits are provided in parentheses. Further details about the methods are provided in the main text.

- Results obtained through the use of double-orthogonalization and any sensible ML method for estimating the “nuisance functions” should be similar.
- 5-fold cross fitting yields smaller SEs (using more observations to estimate ML) than 2-fold.